EC2 Hands-on Assignment

Objective

Learn how to launch, configure, scale, and deploy a simple web application using AWS EC2.

Scenario

You are a Cloud Engineer. Your task is to deploy a simple web server on AWS EC2 and scale the instance resources as needed.

Requirements

1. Launch EC2 Instance

• Use Amazon Linux 2 or Ubuntu
• Instance type: t2.micro or t3.micro
• Create a new key pair
• Configure Security Group:
- SSH (22) → Your IP
- HTTP (80) → Anywhere (0.0.0.0/0)

2. Connect to EC2

```
ssh -i your-key.pem ec2-user@<public-ip>
# or
ssh -i your-key.pem ubuntu@<public-ip>
```

3. Install Web Server

For Amazon Linux:

```
sudo yum update -y
sudo yum install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

For Ubuntu:

```
sudo apt update
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

4. Deploy a Custom Web Page

```
sudo nano /usr/share/nginx/html/index.html

<h1>Welcome to My EC2 Server</h1>
```

```
<p>Deployed by Your Name</p>
```

5. Test the Application

```
http://<public-ip>
```

6. Scale CPU and RAM (Modify Instance Type)

• Stop the EC2 instance
• Go to Actions → Instance Settings → Change Instance Type
• Select a higher instance type (e.g., t3.small or t3.medium)
• Start the instance again
• Verify application is still running

Submission Requirements

• Screenshot of EC2 instance running
• Screenshot of web page in browser
• Screenshot before and after scaling instance type
• Commands used
• Public IP
• Short explanation (2–3 lines) of what you learned

Learning Outcomes

• Understanding EC2 basics
• Working with security groups
• Installing and managing web servers
• Scaling compute resources (CPU/RAM)
• Basic Linux commands
• Deploying applications on cloud